Finite-Source Cosmology from Observer-Patch Holography

Source, Transfer, and Likelihood Boundaries

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Abstract

We give a conditional source-screen construction for Observer-Patch Holography (OPH). A supplied primitive collar law fixes the angular amplitude, and an edge-center generator with its orientation-half identity fixes the tilt once its reserve-generator receipt is supplied. Physical dilation and radial cross-covariance tomography provide two sufficient uniqueness routes beyond one-shell data. The construction states the source, covariant stress, physical scale, Boltzmann transfer and likelihood conditions needed to convert these screen quantities into cosmological observables. For a supplied layered event population with a conformally flat causal order, a spatially flat expanding background is a record density per comoving cell and conformal tick; native repair selects neither the population nor its profile. The redshift is a fourth root of a count ratio for congruent comoving windows with a constant scale factor within each window, or in the shrinking-window limit of a continuous profile. The receipt-conditional primordial values, the tilt $1 - P_\star/48 = 0.96602$, zero running, zero source-side isocurvature and zero linear primordial tensor amplitude under additionally vanishing tensor initial data and forcing, are compared with Planck 2018, ACT DR6, SPT-3G and BICEP/Keck data that precede the registered freezes, in a deterministic ledger with machine-checked brackets. The tilt lies within one standard deviation of Planck 2018 and of both SPT-3G combinations, and at the lower 0.612 percent position of the official ACT DR6, Planck, lensing and DESI BAO joint posterior chains, a retrospective tension under baseline ΛCDM that is not a likelihood ratio.

1 Claim Boundary

A physical cosmological prediction requires one construction that maps finite OPH records to a covariant source, propagates that source through a declared background, and evaluates the resulting observables with a complete likelihood. No such construction is supplied. The executable observer construction runs native repair, records and local event clocks on one twelve-port carrier and on refining spherical source screens, where native density snapshots have a two-dimensional continuum limit [3, 5]. It supplies no covariant source, physical scale, radial lift or primordial branch, and the conditional tilt is not derived from its joint records [7]. Simulator outputs may test numerical interfaces and expose incompatible assumptions. They carry no cosmological verdict.

The same source premises govern the dark-sector, cosmological-vacuum, and compact OPH constructions.

2 Finite-Source Interface

Let X_r denote a finite observer-screen state at refinement level r. A cosmological source map must emit

$$\mathsf{S}_r(X_r) = (g_{\mu\nu}, T_{\mu\nu}, \mathcal{I}, \mathcal{B})_r, \tag{1}$$

where $T_{\mu\nu}$ is conserved, $\mathcal{I}$ contains gauge-invariant initial data, and $\mathcal{B}$ states the background and boundary conditions. The map must be defined without using the observed sky quantities that its outputs are compared with.

The complete one-shell covariance determines the three-dimensional correlation only for chord lengths $0 \le s \le 2R_\star$. The resulting radial kernel is infinite-dimensional and persists under positivity. Two sufficient routes for a source-derived lift are a scale-natural embedding that transports refinement to physical log-wavenumber dilation and complete radial cross-covariances supporting spherical-Hankel tomography. Each route has its own hypotheses. The physical source map must also specify the radial measure, null-space treatment, clock, normalization, and non-fitting forward residual. No finite source construction satisfying these conditions is supplied here. Its source-dependency graph must be finite and acyclic. This graph is a scientific prerequisite graph among stress, clock, scale, radial, transfer, and likelihood objects. It is distinct from the semantic read-from order of the consensus layer. If its nodes are instantiated by semantic commits, a declared mapping and complete certified read supports can make the generated informational order verify their execution dependencies. The provenance theorem alone derives neither these scientific nodes nor their prerequisite edges. The source Gram construction supplies a rank-three Euclidean metric on one twelve-port carrier and, after adjoining a real axis, an algebraic Lorentzian carrier. A nonzero closed convex pointed displacement cone is forced to be future or past Lorentzian if the proper icosahedral rotations and all rapidities of one boost axis preserve it. This is an operational condition on histories, interventions and clock readouts; an algebraic frame action alone does not satisfy it. A time orientation fixes the future branch [1]. The rank-three metric is a finite readout of that carrier's slow response, not an extended space. On the refining source screens, bounded-energy port modes lose the within-carrier band, including the rank-three band, at order 2^{-l}, and the native continuum generator has heat-trace spectral dimension two on the two-dimensional source [3, 4]. In the native repair history, dependence crosses at most one seam per ordered repair event, a speed bound in graph hops per native event that fixes no laboratory speed or continuum light cone [6].

A conservative Fibonacci-record family realizes a controlled flat $1+3$ limit through a specified population, complete shrinking-radius neighbour reads and layer duration. Its normalized event counts converge to spacetime volume, and interior timelike intervals have ordering fraction tending to $1/10$. Given complete authenticated ancestry and interval anchors, an observer can read $(|I|/|J|)^{1/4}$ relative to a nondegenerate interval J in the same history. It converges to the proper-time ratio without using coordinates, timestamps or density in the readout. The population and local read law are hypotheses of the limit, and the reference interval fixes only a relative time unit [1].

Cosmology requires a common physical realization of those events and clocks with expanding geometry, conserved stress, a primordial source, a radial lift and photon transfer. The finite tensor theorem uses a separately supplied $3+1$ tensor interface, nine algebraic null balances, symmetry, discrete conservation and connectedness to obtain an Einstein-shaped identity with one constant metric term. It does not identify its tensor-coordinate differences with source-event displacements. Smooth Einstein promotion requires same-family tensor-curvature convergence or the continuum small-ball/null-balance identification; scalar-curvature convergence alone is diagnostic. The flat

count-clock construction does not supply these cosmological attachments or a likelihood.

2.1 Expansion on the sourced order

The transfer calculation of Section 6 runs on a spatially flat Friedmann background. On the source family that background has an exact finite form: the layered read law in comoving coordinates is conformally invariant, so the flat causal order of the spacetime construction is the causal order of every spatially flat expanding history, and the scale factor enters only through the count measure, as the weight a^4 per comoving cell and conformal tick that a constant physical event density assigns. The count clock converges to the proper-time ratio for shrinking diamonds; fixed finite diamonds obey a scale-factor enclosure and can retain a continuum bias. The redshift between two epochs is the fourth root of the ratio of record counts of congruent comoving diamonds when the profile is constant within each, or in the corresponding shrinking-window limit. The statements are proved in the fragment below and machine-checked in the named Lean module. The record-production profile $a(\eta)$ is a supplied datum on this family; deriving it from the source dynamics is the sourced-density problem of the spacetime paper, and nothing here supplies the Friedmann equation, a physical clock, or the physical scale map. In the fragment, conformal time η, the layer index j and the durations $\sigma_j\Delta$ called physical ticks are coordinates of the supplied FLRW chart; none is a calibrated observer clock.

2.2 Expansion as record density on the source order

The spatially flat Friedmann–Lemaître–Robertson–Walker metric in conformal time η is

$$ds^2 = \sigma(\eta)^2\big(c^2\,d\eta^2 - |dx|^2\big), \tag{2}$$

with scale factor $\sigma(\eta) > 0$; the letter a is the edge radius of the read law. This metric is a positive conformal factor times the flat metric in the coordinates (η, x). A positive conformal factor leaves the causal order of a Lorentzian metric unchanged, and the causal order fixes the metric up to such a factor (Hawking, King and McCarthy 1976; Malament 1977). On the spatially flat family the conformal factor is the entire background, so every profile σ has the causal order of flat space in comoving coordinates. The layered read law of the source-net construction places an event at every (j, s) for a site s in a population S, lets $(j+1, t)$ read (j, s) whenever $\|t - s\| \le a$, and takes the reflexive transitive closure of the reads as the order. Written in comoving coordinates with the tick $\Delta = a/c$ in conformal time, this law generates one finite order for every profile. The profile enters only through the count measure.

Proposition 2.1 (Conformal invariance of the layered order)**.** *Let S be a subset of a real normed space, $a > 0$ an edge radius and $\sigma_j > 0$ the scale at layer j. The physical position of the site s at layer j is $\sigma_j s$ and the physical read radius at that layer is $\sigma_j a$. The physical read from (j, s) to $(j+1, t)$,*

$$\|\sigma_{j+1}t - \sigma_{j+1}s\| \le \sigma_{j+1}a,$$

holds exactly when $\|t - s\| \le a$. The layered orders generated by the physical reads and by the comoving reads coincide, for every profile $(\sigma_j)_{j\ge 0}$.

Proof. For $\sigma > 0$, $\|\sigma t - \sigma s\| = \sigma\|t - s\|$, and multiplication by σ preserves and reflects the comparison with σa. The two read relations are equal, hence so are their reflexive transitive closures. □

Proposition 2.2 (FLRW count measure)**.** *Assign a comoving cell of volume v_s to every site s and the conformal tick Δ to every layer. Under a constant physical event density ρ, the physical tick at layer j is $\sigma_j\Delta$ and the physical cell volume is $\sigma_j^3 v_s$, so a finite set I of events carries the mass*

$$M_\sigma(I) = \rho \sum_{(j,s)\in I} (\sigma_j\Delta)(\sigma_j^3 v_s) = \rho \sum_{(j,s)\in I} \sigma_j^4\, \Delta\, v_s, \tag{3}$$

the discrete form of the four-volume weight $\sigma^4\, d\eta\, d^3x$. Write $M_1(I) = \rho\Delta \sum_{(j,s)\in I} v_s$ for the flat mass. If $0 \le \sigma_{\min} \le \sigma_j \le \sigma_{\max}$ on the layers of I, then

$$\sigma_{\min}^4 M_1(I) \le M_\sigma(I) \le \sigma_{\max}^4 M_1(I). \tag{4}$$

A constant profile $\sigma_j = \sigma$ gives $M_\sigma(I) = \sigma^4 M_1(I)$.

Proof. The identity in (3) is the product of the two scaled factors. Fourth powers are monotone on nonnegative reals and $\Delta v_s \ge 0$, so each term of the sum lies between the same term with σ_j replaced by $\sigma_{\min}$ and by $\sigma_{\max}$. Summing proves (4); the constant case is the common factor σ^4. □

Corollary 2.3 (Count clock in an expanding family)**.** *Let I and J be finite event sets with uniform cells, J nonempty, with $\sigma^I_{\min} \le \sigma_j \le \sigma^I_{\max}$ on the layers of I, $\sigma^J_{\min} \le \sigma_j \le \sigma^J_{\max}$ on the layers of J, $\sigma^I_{\min} \ge 0$ and $\sigma^J_{\min} > 0$. Then*

$$\frac{\sigma^I_{\min}}{\sigma^J_{\max}} \left(\frac{|I|}{|J|}\right)^{1/4} \le \left(\frac{M_\sigma(I)}{M_\sigma(J)}\right)^{1/4} \le \frac{\sigma^I_{\max}}{\sigma^J_{\min}} \left(\frac{|I|}{|J|}\right)^{1/4}. \tag{5}$$

Let I and J approximate comoving diamonds of conformal durations T_I and T_J in the sense of the count-clock theorem for the flat order, so that $(|I|/|J|)^{1/4} \to T_I/T_J$ under refinement. A comoving worldline through a diamond of conformal duration T accrues proper time between $\sigma_{\min}T$ and $\sigma_{\max}T$. In the refinement limit the clock read from the physical masses, $(M_\sigma(I)/M_\sigma(J))^{1/4}$, and the proper-time ratio τ_I/τ_J lie in one interval whose endpoints have the quotient $(\sigma^I_{\max}/\sigma^I_{\min})(\sigma^J_{\max}/\sigma^J_{\min})$; the two readings agree up to this factor in either direction. For a profile continuous in η and diamonds shrinking to points, the factor tends to one and the clock reads the proper-time ratio exactly in the limit.

Proof. Divide the two sandwich inequalities (4) for I and J, with $M_1(I)/M_1(J) = |I|/|J|$ for uniform cells, and take the increasing fourth root; the fourth root of σ^4 is σ. The proper time along the comoving worldline is $\int \sigma\, d\eta$ over the conformal duration, bounded by the extreme scales times T; the ratio of two such enclosures has the stated endpoint quotient. Continuity of σ forces $\sigma_{\max}/\sigma_{\min} \to 1$ on shrinking diamonds. □

Proposition 2.4 (Finite expanding diamonds need not read proper time)**.** *For the supplied radiation profile $\sigma(\eta) = 1 + \eta$ and comoving vertical diamonds starting at $\eta = 0$, take conformal durations 1 and $1/2$. Their continuum mass ratio is*

$$\frac{M_\sigma(1)}{M_\sigma(1/2)} = \frac{1516288}{44451} \ne \left(\frac{12}{5}\right)^4 = \left(\frac{\tau(1)}{\tau(1/2)}\right)^4.$$

Thus their fourth-root volume clock differs from the proper-time ratio even after arbitrarily fine discretization at these fixed durations.

Proof. The common factor $4\pi\rho c^3/3$ cancels from the two integrals

$$M_\sigma(T) \;\propto\; \int_0^{T/2} (1+\eta)^4 \eta^3 \, d\eta + \int_{T/2}^{T} (1+\eta)^4 (T-\eta)^3 \, d\eta.$$

Expanding the polynomials and integrating gives the displayed rational ratio. Along the comoving worldline, $\tau(T) = \int_0^T (1+\eta)\, d\eta = T + T^2/2$, hence $\tau(1)/\tau(1/2) = 12/5$. Subtracting fourth powers gives the strictly positive rational $25944064/27781875$. □

`Lean/Geometry/SourceNetClockSeparation.lean` checks the evaluated mass polynomial and rational mismatch; the identification of that polynomial with the geometric integral is the analytical argument above. The resulting relative clock bias is $+0.696366\%$. It is a continuum finite-window effect, not a discretization error. For the supplied profiles $\sigma(\eta) = (1-\eta/2)^{-1}$ and $\sigma(\eta) = [1 + (\sqrt{2}-1)\eta]^2$, independent split quadratures give -1.595321% and $+0.265422\%$ at the same durations. The shrinking-diamond hypothesis in Corollary 2.3 is therefore substantive. The exact polynomial control and numerical receipts accompany `code/observer_dynamics/verify.py`. No expansion profile is selected by this calculation.

Corollary 2.5 (Redshift as a count ratio)**.** *Let two congruent comoving diamonds, with the same comoving cells and the same conformal ticks, lie at the emission epoch, of scale σ_e, and at the reception epoch, of scale σ_0, the profile being constant across each. Their masses are $n_e = \kappa\sigma_e^4$ and $n_0 = \kappa\sigma_0^4$ with the common factor $\kappa = \rho\Delta \sum_{(j,s)\in I} v_s$, and*

$$1 + z = \frac{\sigma_0}{\sigma_e} = \left(\frac{n_0}{n_e}\right)^{1/4}. \tag{6}$$

Proof. The constant case of Proposition 2.2 gives both masses; κ cancels in the ratio, and the positive fourth root of $(\sigma_0/\sigma_e)^4$ is σ_0/σ_e. □

Remark 2.6 (Expansion history as a record-production law)*.* On this family the Friedmann equation is a statement about record production: a comoving cell of volume v produces $\rho\,\sigma(\eta)^4\,\Delta\, v$ records per conformal tick, and a profile $\sigma(\eta)$ is the same datum as a record-production rate per comoving cell. The de Sitter profile $\sigma(\eta) = -1/(H\eta)$ on $\eta < 0$ solves $d\sigma/d\eta = H\sigma^2$, and its record density per comoving cell is $n(\eta) \propto (H\eta)^{-4}$; the fixed-capacity branch of the corpus, $w = -1$, corresponds to this profile. The Lean module `Lean/Geometry/SourceNetConformalRecordDensity.lean` checks the order invariance, the mass identity and sandwich, the clock enclosure (5), the redshift identity and the de Sitter derivative and density identities over the reals.

Three items lie outside this construction. The source law selecting the profile, the record-production rate per comoving cell, is not derived; the profile is a supplied datum. A reweighted flat order determines at most the conformal factor; on the spatially flat family that factor is the entire background, and spatial curvature $k \neq 0$ changes the comoving spatial metric, which this construction does not treat. No physical clock and no population selection by native repair follow from these identities.

2.3 Conditional source spectrum

On a source-stress-complete uniform-density release cut, the screen field is

$$q_r = \Pi_{\geq 2,r} \left[\frac{1}{3} \log \frac{J_{X,r}}{\bar{J}_{X,r}}\right]. \tag{7}$$

The primitive collar ledger selects a relative-MaxEnt release law before the screen energy is evaluated. It excludes spectra, residuals, likelihoods, fitted amplitudes, and measurement-calibrated proxies. The release energy then fixes

$$A_{q,r} = \frac{1}{d_r}\mathbb{E}_{\nu_r^{\mathrm{rel}}}[q_r^{\mathsf{T}} M_r K_{\theta,r} q_r] = \frac{2E_{q,r}^{\mathrm{src}}}{d_r}. \tag{8}$$

The conformal precision and angular spectrum are

$$K_\theta = \frac{\Gamma(\sqrt{-\Delta_{S^2}+1/4}+3/2+\theta/2)}{\Gamma(\sqrt{-\Delta_{S^2}+1/4}-1/2-\theta/2)}, \tag{9}$$

$$C_\ell^q = A_q \frac{\Gamma(\ell-\theta/2)}{\Gamma(\ell+2+\theta/2)}, \qquad \ell \geq 2. \tag{10}$$

The round-sphere Laplacian $-\Delta_{S^2}$ is a premise of this spectrum. The native continuum generator A of the source screen has energy between fixed multiples of the gradient energy of the flat comparison polyhedron; its identification with $-\Delta_{S^2}$ is not established [4]. A required infinitesimal reserve-generator receipt must supply a full-collar density $P_\star/24$. A separately proved weighted orientation-reversal coarea identity then supplies its source-facing half, so

$$\theta = \frac{P_\star}{48}, \qquad n_s = 1 - \frac{P_\star}{48}, \qquad \kappa_{\mathrm{rep}}^{\mathrm{edge}} = \frac{P_\star}{48(P_\star - \varphi)}. \tag{11}$$

A finite one-step survival probability has exponent $-\log u_q(\log b)/\log b$ and does not supply this infinitesimal density.

On the source-dilation branch, the finite source embedding must satisfy

$$D_{s,r}U_r = U_r R_{s,r}, \qquad C_{\zeta,r} = U_r C_{\mathrm{src},r} U_r^*. \tag{12}$$

For a tilted spectrum on all positive wavenumbers, require convergence on the common compact-$\ln k$ core, with the local dilation residual tending to zero, as specified by the source-orbit-transfer theorem of the radial-lift packet in the spacetime paper [1]. A global uniform covariance norm bound would be incompatible with nonzero tilt and unitary dilation. The local receipt instead gives, on that core,

$$D_s^{-1} C_\zeta D_s = e^{-\theta s} C_\zeta. \tag{13}$$

For a nonzero nonnegative radial covariance, this identity forces

$$\Delta_\zeta^2(k) = A_\zeta (k/k_\star)^{-\theta}, \qquad A_\zeta = \frac{A_q}{\pi^{3/2} Z_q^2 (k_\star R_\star)^\theta} \frac{\Gamma(3/2+\theta/2)}{\Gamma(1+\theta/2)}. \tag{14}$$

The tomography branch obtains uniqueness from complete radial cross-covariances. A finite prior selects a conditional radial continuation. Every finite branch publishes its singular spectrum, null basis, finite-window error, positivity check, and forward residual. Temperature and polarization spectra belong to the transfer interface.

2.4 What the native observer records determine

The fair balanced integer repair law supplies an absolute spectrum for a specified load readout, without fitting an observed sky. On a connected weighted port graph with p ports and r raised

loads, stationary continued repair, in the continuous native attempt parameter t at unit attempt rate per seam, gives

$$\mathrm{Cov}(z(t), z(0)) = \kappa e^{-|t|L/2}\Pi, \qquad \kappa = \frac{r(p-r)}{p(p-1)}, \quad \Pi = I - \mathbf{1}\mathbf{1}^{\mathsf{T}}/p.$$

For the ordinary local average $f_T = T^{-1}\int_0^T z(t)\,dt$,

$$\mathrm{Var}(v_\lambda^{\mathsf{T}} f_T) = \kappa \frac{2(u - 1 + e^{-u})}{u^2}, \qquad u = \lambda T/2.$$

The fixed-graph fluctuation record $T^{-1/2}\int_0^T z(t)\,dt$ instead converges to a Gaussian law with covariance $4\kappa L^+$. Incident pre-attempt records, including null attempts and centered with the prepared filling, have the same leading inverse-Laplacian term with operator correction at most $3\kappa/d_{\min}$, at a common native attempt parameter; t is not a calibrated observer clock. These are analytical consequences of actual local port interaction and readback, checked by finite-state enumeration and independent moment evolution. The detailed proofs and the distinct occupation and event normalizations are given in the screen-microphysics companion [2].

The amplitude is in native load-record units. Occupancy, attempt clock, duration, gain and stationary preparation are specified inputs. For a centered geometric readout $q = Bz + g$, orthogonal projection in the stationary law gives at every finite duration

$$C_T(q) = BC_T(z)B^{\mathsf{T}} + C_T(g), \qquad C_T(g) \succeq 0.$$

Thus a future measured collar-volume readout can test its density overlap; the load record cannot be substituted for that geometric observable. Along one common subsequence of refining source screens with $240 \cdot 4^l$ ports, one-third filling from the declared uniform fixed-number preparation, independent uniform seam selection and fair repair, normalized native occupation snapshots read at native attempt count $\lfloor 1260 \cdot 16^l \tau \rfloor$ converge jointly to a stationary Gaussian family with covariance $\langle f, e^{-|\tau-u|A} g\rangle$. Here A is one positive self-adjoint generator whose form is a conservative, regular, strongly local Dirichlet form of heat-trace spectral dimension two [4, 5]. These finite-dimensional laws assert no path-space tightness or spatial convergence rate, and the density field lives on the two-dimensional screen readout; it is not identified with a primordial curvature field. With fixed-area equal port weights and a common deterministic filter of bounded L^1 gain W, the ensemble mean empirical harmonic pseudo-power satisfies $\widetilde{C}_\ell = (2\ell+1)^{-1}\sum_m \mathbb{E}|a_{\ell m}|^2 \le \pi W^2/(p-1)$, even without exact quadrature. This is not a pointwise bound on every sky realization. Ordinary averages have $W = 1$; amplified fluctuation records do not have uniformly bounded gain as their duration grows.

There are two additional quantitative restrictions. A common deterministic temporal filter of stationary density obeys $d\log V/d\log\lambda \ge -1$, so under a separately established three-dimensional $\lambda \propto k^2$ identification it cannot produce the required k^{-3} curvature spectrum. Under the same operator dictionary that identification fails for the native continuum generator: its eigenvalues obey $n/49920 \le \lambda_n(A) \le 3n/2$, which excludes a unitary identification, up to a fixed scale and bounded mass shift, with a second-order elliptic operator on a smooth compact three-dimensional manifold, and excludes a boundedly invertible energy congruence with the static boundary action of a smooth compact three-dimensional Dirichlet bulk [4]. The native port operator is also not proved round: at level four the seven-mode band splitting is 14.642%, compared with 0.01561% for a supplied round-metric operator. An inverse graph covariance retains its precision's anisotropy.

Absolute inherited/fresh/settled covariance has been recovered for all 61 retained refinement chains and eight group sizes. For unshuffled level-eight to level-nine chains at 65,536 carriers per group, the inherited-only residual variance is $2.55666955 \times 10^{-6}$; removing the linear fresh-load response leaves $4.00020779 \times 10^{-8}$. The first variance times group size is 0.16755, close to the supplied fresh variance 1/6. Nested same-location residuals satisfy $\mathrm{Cov}(e_M, e_N) = \mathrm{Var}(e_{\max(M,N)})$; this is a projection identity, not a separated-position covariance or independence theorem. Group-dependent inherited responses do not supply a common refinement generator. The retained sample shares initial fields, and fresh reconstruction depends on pinned producer and random-number semantics; it supplies neither A_s nor n_s. The specifications, original chain receipts, independent controls and absolute-covariance extraction are archived [12].

2.5 What a fixed-geometry repair history identifies

The repair carrier has locally readable port loads, seam attempts and records. Its supplied cell chart and incidence are not updated by a load repair. This distinguishes two possible source constructions: reading the geometry present in that implementation, and attaching a proposed volume law to its load history.

Proposition 2.7 (Frozen geometric readout)**.** *Let a state be (Γ, x), and suppose every allowed repair has the form $(\Gamma, x) \mapsto (\Gamma, F_e(x, \xi))$. For any positive cell-volume readout $V_i(\Gamma)$, prepared reference $\bar{V}_i = V_i(\Gamma)$, and finite repair history,*

$$q_i = \frac{1}{3} \log \frac{V_i(\Gamma)}{\bar{V}_i} = 0.$$

Every unequal-position covariance and every linear coarse/fine covariance of this field is zero, under every load preparation, schedule and recording clock. The conclusion applies separately at every refinement level.

Proof. Induction over the history preserves the first component. Its readout equals the reference pointwise, so the log contrast vanishes before any expectation or coarse graining is taken. □

This proposition concerns perturbations of the supplied geometry. It says neither that the background geometry is flat nor that all possible observer dynamics have zero curvature. It also does not prohibit an operational metric reconstructed by a new constitutive rule from load-dependent probes. Such a rule would add the missing metric-volume interface.

On the native geodesic carrier, a concrete chart diagnostic is the positive cone volume $V_i = |\det(v_{i0}, v_{i1}, v_{i2})|/6$ of a triangular cell. It is an imported Euclidean chart volume, not a measured physical collar volume. The finite controls apply the production integer repair to both a uniform half-filled preparation and a clustered half-filled preparation at 20 and 80 cells, with 240 and 960 ports. All 64 attempts in each control are retained and independently replayed. Rotating the chart preserves these volumes; scaling all coordinates by two multiplies them by eight. Both operations leave the log contrast with the matched reference zero. Comparing absolute cell sizes at different refinements does not turn this background change into a stochastic perturbation.

Proposition 2.8 (Equal local variance does not identify source shape)**.** *Consider the isolated twelve-port carrier's internal icosahedral Laplacian L, $\Pi = I - \mathbf{1}\mathbf{1}^{\mathsf{T}}/12$, and native stationary fixed-occupancy load z. Both z and the local seam drive Lz are locally computable, mean-zero, and*

equivariant under carrier rotations. After separately normalizing their covariance traces to one, they give

$$C_{\mathrm{load}} = \frac{\Pi}{11}, \qquad C_{\mathrm{drive}} = \frac{L^2}{360}.$$

Both diagonals are $1/12$*. At antipodal ports their covariances are* $-1/132$ *and* 0*, respectively; at adjacent ports they are* $-1/132$ *and* $-1/45$*. Thus locality, carrier symmetry, mean removal and equal one-point variance leave separated-position power unselected.*

Proof. Native stationarity gives $\mathrm{Cov}(z) = \kappa\Pi$. Since $L\mathbf{1} = 0$, the drive covariance is κL^2. The graph is five-regular, so $(L^2)_{ii} = 30$ and $\mathrm{tr}(L^2) = 360$. Adjacent vertices have two common neighbors, yielding $(L^2)_{ij} = -10 + 2 = -8$; antipodal vertices have neither an edge nor a common neighbor, yielding zero. Division by the respective traces proves the statement. Equivariance follows because every carrier automorphism commutes with L. □

This is the exact internal seam graph used by native repair and by its isolated-carrier control; inter-carrier seams are excluded from this twelve-port calculation. Its covariance matrices are not asserted to be the full coupled federation covariances. The filling, stationary reference and gains used to normalize the traces are declared inputs. The normalization is a diagnostic comparison, with no observed spectrum used to choose either gain. Exact enumeration retains both the one-raised and six-raised stationary preparations, with 12 and 924 configurations and $\kappa = 1/12$ and $3/11$. It checks all covariance entries, including two disjoint antipodal two-port blocks and their fine/coarse cross covariance within that carrier. This block comparison is not a cosmological separation or a continuum refinement limit. For a centered current-state readout, $q = Bz + g$ with $B = \kappa^{-1}\mathbb{E}[qz^{\mathsf{T}}]$, the same calculation retains the absolute density component and residual. The local mismatch readout $\sum_{j\sim i}(n_i - n_j)^2$, centered under its preparation, has $B = 0$ and a nonzero residual at half filling: complementing every load leaves this readout invariant while reversing z. Nonlinearity therefore supplies an additional load observable, but by itself no geometric meaning.

The exact record controls use $P = I + G/30$, with the continuous generator G at unit attempt rate per seam: one tick is a uniformly selected seam attempt with a fair endpoint coin. For $m = 1, 2, 4$ retained readings and $c = 1, 2$ attempts between readings, they check

$$\mathrm{Cov}\left(\frac{1}{m}\sum_{a=0}^{m-1} q(X_{ca})\right) = \frac{mC_0 + \sum_{a=1}^{m-1}(m-a)(C_{ca} + C_{ca}^{\mathsf{T}})}{m^2}, \qquad C_t = \mathbb{E}[q(X_0)q(X_t)^{\mathsf{T}}].$$

These altered clocks change record covariances and leave the supplied geometry invariant. They are finite exact checks, not a physical time calibration or an asymptotic refinement experiment.

For any of these load observables u_i, one may attach positive labels $J_i/\bar{J}_i = \exp(3u_i)$, giving a formal log-volume contrast u_i without changing a single native load transition. These are unselected extensions, not certified physical models. Even gain calibration cannot choose between the distinct covariance shapes above. A nontrivial geometric source needs an actual metric or collar-volume update, its relation to native records, an independently specified reference slice, and a physical calibration. The fixed-geometry construction supplies none of those by relabelling a load fluctuation. Nonstationary preparations, spatial memory or new couplings are possible mechanisms, but each must expose that interface.

Reconstructing a chart solely from the exact native linear response does not introduce a stochastic geometry update: the fair repair law has $Gx = -Lx/2$ with the same prescribed L. Any fixed

reconstruction from this operator is independent of the realized load preparation. A state-dependent metric extracted from other response or fluctuation data would require its own observable and constitutive identification.

The general finite-history and observability statements, together with a four-cycle equal-diagonal covariance witness, are kernel-checked in the Lean geometry library.[1] The twelve-port rational enumeration and native trace checks are reproduced by an independent verifier with retained inputs and outputs.[2] No primordial amplitude, tilt, physical expansion or photon transfer is inferred from these controls.

2.6 Exact Fourier control for the golden source population

The deterministic Cartesian population $x_b = \mathcal{L}(\{b_1\varphi\}, \{b_2\varphi\}, \{b_3\varphi\})$, $0 \le b_i < q$, has an exact fixed-cube Fourier response. Here $\varphi = (1+\sqrt{5})/2$, $\mathcal{L}$ is the declared cube size, and the integer mode n has wave vector $2\pi n/\mathcal{L}$. Its normalized amplitude factors as

$$F_q(n) = q^{-3} \sum_b e^{2\pi i n\cdot x_b/\mathcal{L}} = \prod_{j=1}^{3} a_q(n_j), \qquad a_q(0) = 1,$$

$$a_q(m) = e^{\pi i m\varphi(q-1)} \frac{\sin(\pi m q\varphi)}{q\sin(\pi m\varphi)} \quad (m \neq 0).$$

This follows by separating the three finite sums, using integer-mode periodicity to remove fractional parts, and evaluating each geometric series. For $q = F_j$, the Fibonacci identity $\varphi F_j - F_{j+1} = (-1)^{j+1}\varphi^{-j}$, with $F_j\varphi^{-j} \to 1/\sqrt{5}$, gives for fixed n with h nonzero coordinates

$$q^{4h}|F_q(n)|^2 \longrightarrow \prod_{n_a\neq 0} \frac{(\pi n_a)^2}{5\sin^2(\pi n_a\varphi)}.$$

The denominator is nonzero by irrationality. The small-angle limit of each numerator proves the rate. It concerns fixed integer modes in a fixed cube. It is not an infinite-volume hyperuniformity result or a primordial tilt. Finite directional power ratios can be large while both powers vanish; some finite modes remain above the independent-uniform structure-factor reference. The exact formula, independent small-cloud checks and the retained numerical receipt are archived [11]. No physical length or curvature amplitude is selected by this supplied coordinate population. The Lean module `Geometry.GoldenSourceFourier` [10] checks tensor factorization, fractional-coordinate phase equality and the finite geometric-series identity. The fixed-mode asymptotic argument above is analytical, not a claimed formal limit theorem.

2.7 Common-bulk consistency beyond a single screen

A further sufficient uniqueness route uses an unbounded sequence of round radii, on one common Euclidean time slice and in one fixed normalization. If one homogeneous isotropic generalized bulk covariance has exact screen spectra $C_\ell(R_j) = 2\pi A_j/[\ell(\ell+1)]$ for all $\ell \ge 1$, then overlapping

[1] https://github.com/FloatingPragma/observer-patch-holography/blob/r2051/Lean/Geometry/NativeGeometricSourceIdentification.lean

[2] https://github.com/FloatingPragma/observer-patch-holography/tree/r2051/code/native_geometric_source

chord intervals force all $A_j = A$, $\xi(s) = -A \log s + c$ almost everywhere, and $P_3(k) = 2\pi^2 A/k^3$ for $k > 0$. The radial theorem assumes well-defined restrictions, a locally integrable off-diagonal covariance and no contact terms. Removing the dipole retains $c + ds^2$ freedom. A finite family of separate screen marginals does not supply the common bulk or unbounded coverage: positive perturbations supported beyond the largest measured chord preserve all those marginals. The detailed radial theorem and its explicit positive-baseline control therefore strengthen the conditional inverse statement without identifying the native load record as curvature. The native screen energy supplies no such common bulk: by the boundary-congruence exclusion of Section 2.4, a bulk reading of the screen needs a radial observation algebra that the native energy does not supply [4].

3 Dark-Sector Interface

The dark-matter paper proposes anomalous modular charge on overlap collars. For a conserved total charge Q_A in a comoving region of physical volume $V_0 a^3$, its homogeneous density satisfies

$$\rho_A = \frac{Q_A}{V_0 a^3} \propto a^{-3}. \tag{15}$$

Under the separate homogeneous continuity equation with no exchange, this dilution is the pressureless background law. Deep-regime scale covariance and quadrature composition characterize the static spherical acceleration law and baryonic Tully–Fisher scaling.

Equation (15) does not determine the abundance, initial perturbations, relativistic stress, gravitational slip, sound speed, nonlinear evolution, or coupling to the visible sector. A cosmological use requires a covariant completion that supplies these quantities from one common source and obeys the full energy-momentum ledger. That completion is not supplied here.

4 Primordial Interface

A physical primordial branch must supply all of the following:

1. a finite source ensemble and a clock;
2. a conserved covariant stress tensor;
3. gauge-invariant scalar, vector, and tensor initial data;
4. a screen-to-volume lift with a controlled null space;
5. phase, amplitude, and isocurvature conditions;
6. refinement stability and an error model.

The conditional source theorem selects $n_s = 1 - P_\star/48$ from the infinitesimal edge-center generator receipt. Physical primordial use requires one finite source construction that supplies this receipt together with the release law, stress, clock, freezeout, mode, scale, and radial receipts. Numerical agreement with a measured summary does not supply the missing source map.

5 Comparison with Public Data

The values that the conditional theorems of this paper and its companions fix without a dial are compared with public measurements in a deterministic ledger released with these papers, with an independent verifier and exact rational brackets machine-checked in the Lean module `CosmologyLedgerBrackets`. Every comparison is on seen data and carries a class label; the rule, theory minus measurement in units of the quoted standard deviation, is fixed before any row is read. The primordial rows are the following.

- The tilt $n_s = 1 - P_\star/48 = 0.96602$ at the comparison pixel, with a branch spread below 10^{-7}, lies 0.27σ above Planck 2018 (0.9649 ± 0.0042), 0.69σ above SPT-3G with Planck (0.9636 ± 0.0035), 0.79σ below the SPT, Planck and ACT combination (0.9684 ± 0.0030) and 1.28σ below ACT DR6 with Planck (0.9709 ± 0.0038). The official joint posterior chains of ACT DR6 with Planck, lensing and DESI 2024 baryon acoustic oscillations, whose mean is 0.9743 ± 0.0034, place the branch at their lower 0.612 percent position and the historical clock alternative $1 - e(P_\star - \varphi) = 0.96484$ at 0.200 percent; the measured tail differs from the Gaussian summary [9, 7]. Those data precede the FZ-01 and FZ-15 registrations, so this is a retrospective tension under baseline ΛCDM, not a likelihood ratio, and the registered comparison with a later dataset is unchanged. The value is a consequence of the reserve-generator receipt, which is not supplied; the comparison records where a receipt-conditional number lands and selects no branch.

- The running vanishes exactly on the dilation-cocycle branch; Planck 2018 gives -0.0045 ± 0.0067 (0.67σ) and ACT DR6 with Planck, lensing and baryon acoustic oscillations gives 0.0062 ± 0.0052 (1.19σ).

- A vanishing emitted transverse-traceless source does not determine the primordial tensor amplitude. The linear tensor equation admits homogeneous solutions: for example, in radiation domination $h_k'' + 2h_k'/\tau + k^2 h_k = 0$ has the nonzero regular solution $h_k = \sin(k\tau)/(k\tau)$. The additional conditions of zero initial tensor amplitude and velocity, and no tensor forcing, select the zero linear solution and hence $r = 0$ when the scalar power is positive. Those tensor initial conditions are not derived by the scalar screen construction. BICEP/Keck 2018 gives $r_{0.05} < 0.036$ and the fourth Planck release with BK18 and BAO gives $r < 0.032$, both compatible with this more restrictive branch. A future null excludes only specified models whose tensor signal exceeds its bound; single-field slow roll has no universal positive lower bound on r. An observable comparison must also separate linear primordial tensors from later scalar-induced tensors, lensing, and foregrounds.

- The source-side isocurvature fraction vanishes on the rank-one single-clock branch; Planck 2018 bounds the uncorrelated cold-dark-matter fraction at 3.5, 3.8 and 3.9 percent at the three reference scales. Zero non-Gaussianity at the source (Planck local -0.9 ± 5.1) and zero curvature (Planck with BAO 0.0007 ± 0.0019; DESI DR2 with CMB 0.0023 ± 0.0011, 2.1σ) are shared with the adiabatic single-field baseline.

The first background row is the fixed-capacity equation of state $(w_0, w_a) = (-1, 0)$, which the Gaussian moment diagnostic of the official DESI DR2 chains places between 2.6σ and 3.9σ from the posterior means of the four default combinations, a tension recorded with the same prominence as the agreements. This fixed-capacity point coincides with ΛCDM, so its survival earns no distinctive OPH support, and the registered FZ-13 rule uses empirical posterior quantiles rather than

a Gaussian conversion [8]. The other background rows are the capacity candidates 3.29×10^{122} and 3.30×10^{122}, whose dictionary $\Lambda \ell_P^2 = 3\pi/N$ returns the cosmological constant within 0.6 percent of the Planck-central value and 2.7σ below the DESI DR2 with CMB display. The conditional tilt protocol FZ-15 and the fixed-capacity protocol FZ-13 were frozen on 2026-10-04; the register labels both `frozen_stamped_upgrade_pending`. Only their complete registered protocols and anchoring requirements govern eligible comparisons. The FZ-15 registered comparison concerns a specified later Simons Observatory product; the Planck, ACT DR6, SPT-3G and DESI data above precede both FZ-01 (2026-07-17) and FZ-15 [8]. The tensor-zero FZ-16 and running/isocurvature FZ-17 texts are unadopted proposals, not scoring contracts. Their historical proposal body is retained separately from the operative FZ-15 target.

6 Transfer Interface

Given a completed source map, the transfer calculation must declare the background species, equations of state, collision terms, recombination model, initial modes, gauge convention, numerical tolerances, and observable projection. Standard Einstein–Boltzmann software can test this interface with imported inputs. Such a run tests the solver path rather than an OPH source theorem.

The minimum transfer outputs are

$$(C_\ell^{TT}, C_\ell^{TE}, C_\ell^{EE}, C_L^{\phi\phi}, P_m(k,z), H(z), D_A(z), f\sigma_8(z)). \tag{16}$$

All outputs must arise from one compatible background and perturbation system.

7 Likelihood Interface

A data comparison must state the datasets, masks, covariances, nuisance parameters, priors, and combination rule. Overlapping samples and shared calibrations must be represented in the covariance model. Visual overlays, diagonal residual sums, and comparisons made after selecting a formula are diagnostics.

The cosmic microwave background (CMB), galaxy, and background comparisons are diagnostic. These computations supply neither a prospectively fixed cosmological comparison nor a falsification verdict.

8 Theorem scope

Object	Classification
Finite observer-screen carriers	Conditional finite construction
Native screen continuum generator A	Proved along one common refinement subsequence; two-dimensional; no physical, radial or primordial identification
Screen scalar covariance	Conditional theorem
Source-functional screen amplitude and edge-center tilt	Conditional theorem; finite receipt not supplied
Radial-lift mathematics	Conditional theorem packet; one-shell unrestricted inversion is impossible
Physical source-dilation or radial-tomography receipt	Not supplied
Primordial covariant source	Not supplied
Repair-charge cosmological abundance	Not supplied
Relativistic repair stress and slip	Not supplied
Boltzmann initial-state bridge	Not supplied
Joint CMB, baryon acoustic oscillation, lensing, growth, and cluster likelihood	Not supplied
Prospectively fixed cosmological comparison	Not supplied by these diagnostics

9 Conclusion

Finite OPH screen data become cosmology only through a covariant source map, a three-dimensional lift, a conserved stress tensor, compatible transfer equations, and a complete likelihood. These constructions are not supplied here. Existing cosmological calculations are diagnostics and carry no falsification verdict.

Credited prior work

The results of this paper build on the following established work. Each entry names an established result as it is used here and cites its sources.

- Conformal invariance of causal structure and the Malament and Hawking–King–McCarthy reconstruction split into conformal class and volume element [16, 19, 17].
- Finite geometric sums, Cartesian product factorization and Fibonacci approximation to the golden ratio [14, 15].
- Legendre completeness, radial Fourier uniqueness and positive spectral perturbations in the null space of sphere restriction [21, 20].
- Multivariate least squares, covariance Schur complements and nested conditional group projections [13, 18].

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